Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/17/2008
PI Name:	Wu, Xiao-lun Ph.D.		
Project Title:	Full-field Interferometric Measurements of Thickness of Freely-suspended Liquid Films		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid transport phenomena		
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
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	We propose to develop interferometric methods to visualize and measure the film thickness in a two-dimensional flowing soap film. Since the film used can be sometimes several optical wavelengths in thickness, enumeration of the interference fringes resulting from a single wavelength in an interferogram is ambiguous and cannot be used to determine the precise value of the thickness. This difficulty can be overcome if interferograms are acquired simultaneously using several different wavelengths or in combination with infrared absorption measurements. The full-field thickness information on flowing liquid films is useful for understanding stress distribution in the film and how this stress relaxes over time and distance. The technique is also useful for understanding the interaction between turbulence and thickness fluctuations, which under certain conditions can be treated as a passive scalar similar to a dye or a temperature distribution.		

Task Description:	Build a flowing soap film channel that allows independent control of the film thickness and flow speed. Simple fluid flow structures, such as a single vortex or a few vortices created by a bluff body will be examined first. More complicated structures, such as a large array of vortices will be studied afterwards. Until recently most of studies of two-dimensional turbulence in flowing soap films have been concentrated on velocity fluctuations. However, our experience shows that accompanying the velocity fluctuations, the thickness of the film also fluctuates. The dynamic role of the thickness variation is not clear at this point. Theoretically, it is predicted that for the typical velocity of the flow to be least than the sound velocity in the film, the thickness variation is passively convected. On the other hand, when the typical velocity of the flow is greater than the sound velocity, the shock waves are created and the thickness variation is no longer passive. The two-dimensional experiment proposed here allows us to investigate the intricate coupling between the velocity field and the flow medium. Similar problems occur naturally in oceans and in atmosphere.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have shown the importance of long chain polymers in the suppression of fluid flows and turbulence. The effect has been previous studied but is not well understood. Our finding points to the significance of hydrodynamic structures, particularly the saddle points in the flow, that are responsible for suppression of turbulence. The finding contributes to our general understanding of polymer effects in flows and may be useful for enhancing fluid transport in pipes and navagation efficiency. The finding that single-point velocity statistics is non-gaussian in 2D turbulence is also intriguing and will motivate new investigations in 3D turbulence.
Task Progress:	We started this project by setting a simple goal of studying fluid flow and turbulence in a thermal driven system, a flowing soap film. A unique feature of turbulence in freely suspended film is that the fluid medium is very thin and the velocity field behaves two dimensionally. Two dimensional (2D) turbulence, aside from its scientific interest, is also relevant to large-scale atmospheric and oceanic flows. Although certain behavior of 2D turbulence is different from its 3D counter part, the commonality of the two systems, such as stochasticity, strong nonlinearity, and self-similarity, is striking, making the study of 2D turbulence valuable. Moreover, the reduced spatial dimension brings about significant simplifications for laboratory studies and theoretical analysis. These simplifying features were exploited in this project. Our initial system was rather small consisting of a square convection cell containing a film that measures 2x2 cm2 and the detecting system was home-built consisting of a single infrared (IR) sensing elements. The project has progressed to the present stage where a film of cylindrical shape, 2 cm tall and 6 cm in diameter, is formed in a uniform vertical temperature gradient delta T. The cylindrical film is continuous in the lateral direction, and thus the flow is homogeneous along that direction. As in most of driven systems, this altered boundary condition introduces additional symmetry to the problem, and new hydrodynamic modes emerge. The new system is also more close to natural convection in the earth atmosphere where flow is unimpeded in the horizontal directions. Thus the main achievements this year include the following: (i) We have designed and built a convection setup that allows a cylindrical film to be established. A new Ph.D. student, Dan Matthews, is mostly responsible for this task. A soap film in the form of an annular ring is stabilized by two cylinders one on the top and the other on the bottom of the film. Both cylinders are temperature controlled and the temperature difference between them can be set manually. The film is slightly pressured from inside so that it resumes a uniform (constant diameter) cylindrical shape; without such pressurization, the film will collapse in the waist, producing undesirable curvatures. The inside pressure is automatically regulated with a laser and a position sensitive diode. To visualize fluid flow, Dan seeded the soap film with a fluorescent dye (fluorescein), which can be exited by a UV lamp and visualized by a CCD camera. Dan did most of the design and machine work; some pieces are made of titanium to prevent corrosion. (ii) With the new setup we have started a series of measurements. We found that the flow patterns in the cylindrical films are indeed very different from that seen in the square film, which we reported early. The difference is the more pronounced when delta T is large so that thermal plumes give way to a large scale flow. In the square films, this large scale flow manifests itself as a circulation that occupies the entire cell. In the cylindrical film, on the other hand, the large scale flow is in the form of shear bands with flow on the top of film moving in one direction while the lower part of the film moves in the opposite direction. The broken left-right symmetry is caused by imperfections in the convection cell and by the initial condition. We found that once the shear band is formed, it is quite stable and can lasts for a long time. Dan also did some programming so that a sequence of images can be analyzed. (iii) We are also working hard to introduce velocity measurements for the circular film. Here we have encounter some difficulties because the curvature of deteriorates the quality of video images, making particle tracking much more difficult than for square films. It appears that we may have to resort to the laser Doppler velocimetry in order to measure turbulent velocity. We have also worked closely with a superb undergraduate student, Rory Cerbus. Rory came to my lab for three months in the summer. We got him interested in the problem of vortex shedding in the soap film. It is an attractive problem for undergraduate because the vortex patterns are very attractive to naked eyes and the physics behind the phenomenon is intriguing. Unlike vortex shedding from a stationary bluff body, such as a cylinder, we are interested in vortex shedding when the body is rotating in the fluid. There are two competing frequencies in the problem, the shedding frequency f of vortices and the rotation frequency fr of the cylinder. We have seen several interesting behaviors as fr increases, ranging breaking up of vortices for small fr to suppression of vortex shedding for large fr. The latter effect may be very useful for reducing wind drags for fast moving objects because vortex shedding is a major factor for friction. This work is nearly completed, and we are looking forward to submit the paper to Phys. Fluids. I am also very glad that Rory is applying for our graduate school, and there is a great opportunity that he will join my group to continue his Ph.D. training at the University of Pittsburgh. As in previous years, we continue to train students at all levels by organizing weekly meetings and discussion sessions. Last year alone we have studied several essential subjects related to fluid mechanics and nonlinear dynamic systems, such as multi-fractal descriptions of turbulence, Kolmogorov theory of turbulence (including a derivation of Kolmotgorov’s 4/5 law), chaos theory, and stochastic differential equations. Students are actively participated in the activity because they are both as students and as teachers; they along with the faculty takes turns presenting the

	materials. This is a very effective way of training student, giving them the opportunity to think the problem through. We also reached out to high school students by running weekend open houses. This year we have accommodated about 10 local high-school students to participate in a workshop to discuss and to perform laboratory experiments in optics. We hope all these activities are useful and encourage them to pursue science and mathematics in the future.
Bibliography Type:	Description: (Last Updated: 05/08/2019)
Articles in Peer-reviewed Journals	Zhang J, Wu XL. "Persistent skewness of a strongly active scalar." Submitted to Phys. Rev. Lett., 2008. , Jan-2008
Dissertations and Theses	Zhang J. "Thermal convection in vertically suspended soap films." Ph.D. thesis, Department of Physics, University of Pittsburgh, 2006. , Jun-2006